

ASHUTOSH ROY

Data Scientist

☎ +91-9810599837 | ✉ ashu2003roy@gmail.com | 🌐 mitovoid.netlify.app | 👤 ashutoshroy02 | 📍 Gurugram, Haryana

SUMMARY

Data Scientist with **1+ year of hands-on experience** in Machine Learning, Data Analysis, Statistical Modeling, NLP, and Predictive Analytics. Published researcher at **RECCAP 2026** with research experience at **IIT Delhi**. Skilled in **Python, SQL, Pandas, Scikit-learn, Power BI, Tableau**, and data visualization. Built ML models and data pipelines that improved prediction accuracy, reduced manual effort, and helped turn large datasets into actionable business insights.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Analyzed and benchmarked **3** OCR models (Tesseract, Marker, Gemini OCR) on **10,000+** archaeological documents, improving extraction accuracy by **22%** and reducing manual effort.
- Built a hybrid search pipeline using **FAISS, Pinecone, and Apache Solr**, improving retrieval accuracy by **35%** across large historical datasets.
- Co-authored the research paper **SARCH: Multimodal Search for Archaeological Archives** with IIT Delhi. arxiv.org/abs/2511.05667

AI/ML Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Built NLP models for healthcare data analysis using feature engineering, improving prediction quality and model performance.
- Developed end-to-end data pipelines for data cleaning, preprocessing, feature extraction, and model inference, reducing manual processing time.
- Improved patient query resolution time by **40%**, increasing support efficiency through better intent classification and response generation.

PROJECTS

Speech Emotion & Stress Detection | Python, PyTorch, Wav2Vec2, BiLSTM, Hugging Face | RECCAP 2026

Jan 2026 – May 2026

- Built a deep learning model for speech emotion and stress detection using **3** public datasets with data cleaning, feature engineering, and preprocessing.
- Achieved **85.9% accuracy**, **0.856 F1**, and **0.1268 RMSE**; improved prediction performance and research accepted at **RECCAP 2026 (IEEE)**.

Netflix Data Analysis | Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter

Jun 2026

- Enhanced an open-source project by cleaning and analyzing data from **8,000+** Netflix titles using Python, Pandas, and NumPy.
- Built visualizations that identified trends in genres, ratings, countries, and release years, helping generate business insights and support data-driven decisions.

Medical Chatbot (LLM Fine-Tuning) | Python, Mistral 7B, QLoRA, Tesseract OCR, Transformers

Mar 2024 – Apr 2024

- Fine-tuned **Mistral 7B** using QLoRA, reducing GPU memory usage by **75%** while maintaining model performance.
- Built an OCR pipeline to process **1,000+** clinical reports, improving data quality and supporting better healthcare predictions.

TECHNICAL SKILLS

Data Science: Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis, Feature Engineering, Machine Learning, Deep Learning, NLP, Regression, Classification, Clustering, Time Series Analysis

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, TensorFlow, PyTorch, XGBoost, Hugging Face

Visualization: Power BI, Tableau, Streamlit, Matplotlib, Seaborn

Databases: SQL, MySQL, PostgreSQL

Languages: Python, SQL, C++

Tools: Jupyter Notebook, Google Colab, Git, Docker, Excel

ACHIEVEMENTS

- Published **2 research papers**, including one accepted at **RECCAP 2026 (IEEE)** and another co-authored with **IIT Delhi**.
- Solved **150+** DSA problems on LeetCode, strengthening data structures and algorithms.
- Won **2 AI/ML Hackathons**, building innovative AI solutions under time constraints.
- Made contributions to **open-source AI/ML projects** on GitHub and Hugging Face.
- Earned certifications in **Machine Learning, Deep Learning, Generative AI, NLP, and Python**.

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh

St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana